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An Experimental -Connectivity Monitoring Metric for IPPM
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Abstract

Within a Segment Routing (SR) domain, segment routed measurement packets can be sent along pre-determined paths. This enables new kinds of active measurements. Connectivity monitoring supervises the state and performance of a connection or a (sub)path from one or a few central monitoring systems. This document specifies an experimental suitable type Type-P connectivity monitoring metric to accommodate such operational needs.

Commenté [MB1]: This is deployment-specific

Commenté [MB2]: For consistency with RFC2680, etc.

Status of This Memo

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Table of Contents

1. Introduction	3
1.1. Requirements Language	5
2. Experiment Goals and Success Criteria	5
3. A segment routing connectivity monitoring framework	6
4. Topology and measurement loop set up requirements	13
4.1. General network topology requirements	13
4.2. Sub-path Monitoring measurement loop routing requirements	13
4.3. Path	15
4.4. Sub-path Monitoring measurement loop packet spacing	15
5. Generic Type-P-SR-Path-Periodic-* metric	15
5.1. Metric Name	15
5.2. Generic Metric Parameters	16
5.3. Metric Units	16
6. Singleton Definition for Type-P-SR-Path- Periodic-Delay	16
6.1. Metric Name	16
6.2. Metric Parameters	16
6.3. Delay Metric Units	16
6.4. Definition	16
6.5. Discussion	17
6.6. Methodologies	17
6.7. Errors and Uncertainties	17
6.8. Reporting the metric	17
7. Singleton Definition for Type-P-SR-Path- Packet-Loss	17
7.1. Metric Name	17
7.2. Metric Parameters	17
7.3. Packet Loss Metric Units	17
7.4. Definition	18
7.5. Discussion	18
7.6. Methodologies	18
7.7. Errors and Uncertainties	18
7.8. Reporting the metric	18
8. Definition of Samples for Type-P-SR-Path-Periodic-Delay	18
8.1. Generic Type-P-SR-Path-Periodic-Delay-* metric	18
8.1.1. Metric Name	18
8.1.2. Metric Parameters	18
8.1.3. Metric Units	19
8.1.4. Metric Defintion	19
8.1.5. Discussion	19
8.1.6. Errors and uncertainties	19
8.2. Definition of Type-P-SR-Path-Periodic-Delay-Stream	19
8.2.1. Metric Name	19
8.3. Definition of Type-P-SR-Path-Periodic-Delay-Variation	19
8.3.1. Metric Name	19
8.3.2. Methodologies	20
8.3.3. Discussion of SRDV	20
8.3.4. Errors and uncertainties	20
8.4. Definition of Type-P-SR-Path-Periodic-Delay-Variation-Stream	20
8.4.1. Metric Name	20
8.4.2. Metric Defintion	20
9. Statistic Definitions for SR-Path-Periodic-*--Stream samples	20
9.1. SR-Path-Periodic-*--Mean	21
9.2. SR-Path-Periodic-*--Std	21
10. Statistic Definitions for Type-P-SR-Path-Packet-Loss	21
10.1. SR-Path-Packet-Loss-Ratio	21

11. Sub-Path monitoring metrics derived from samples captured along the measurement loops	21
11.1. Baseline measurement	22
11.2. Discussion of the baseline measurement	23
11.3. Definition of SR-Path-Sub-Path-RTD-Estimate	23
11.4. Definition of SR-Path-Sub-Path-*--Changepoint	24
11.5. Discussion of SR-Path-Sub-Path-*--Changepoint	25
11.6. Definition of SR-Path-Sub-Path-Congestion-Location	25
11.7. Definition of SR-Path-Sub-Path-Disconnected	26
12. Discussion of Temporal Resolution	29
13. IANA Considerations	29
14. Security Considerations	29
15. References	29
15.1. Normative References	29
15.2. Informative References	31
Author's Address	31

1. Introduction

Within a Segment Routing (SR) domain, measurement packets can be sent along pre-determined segment routed paths [RFC8402]. A segment routed path may consist of pre-determined sub paths, specific router-interfaces, or a combination of both. A measurement path may also consist of sub paths spanning multiple routers, given that all segments to address a desired path are available and known at an SR monitoring tool hosted at an SR domain edge interface, typically.

A Path Monitoring System (PMS) ~~see~~ [RFC8403] is a dedicated central ~~Segment Routing (SR) domain monitoring device~~ function. Monitoring individual sub-paths or point-to-point connections is executed for different purposes. For example, IGP's exchange hello messages between neighbors to swiftly adapt routing after detected topology changes. Network Operators may also ~~have an interest in~~ need to monitoring forwarding, congestion, and connectivity of sub-paths as well as neighbor relationships. Monitoring can be of interest within timescales of seconds, minutes, or hours. The periodicity at which active probing samples are taken and the statistics based on these samples ~~are~~ might be ~~may be~~ more frequent than monitoring of commodity router interfaces based on counters at minute timescale.

The IP Performance Metrics (IPPM) architecture is a first step to that direction [RFC2330]. IPPM's active measurement solutions require dedicated measurement systems, a large number of measurement agents, and synchronised clocks. ~~Edge-Edge-to-to~~ edge domain monitoring by commodity IPPM solutions reduces the total number of required IPPM measurement agents. Localising the site of a detected network anomaly may however require network tomography methods.

Generic ~~IP Performance Metrics (IPPM) metrics~~ are available to monitor connectivity [RFC2678]. These metrics assess connectivity between ~~end~~ nodes without assumptions about the paths connecting them. The metric described

in this document maintains the fundamental definition of connectivity: a monitored sub-path is deemed "available" if no consecutive packet loss occurs. ~~Segment-Routing~~SR enables the design of measurement path setups that support new IPPM metrics and statistics. These metrics and statistics are derived through network tomography applied in a predefined manner, ensuring that repeated measurements precisely produce similar results.

An ~~Segment-Routing~~SR PMS is part of an SR domain. The PMS is IGP topology aware, covering the IP and (if present) the MPLS layer topology [RFC8402] to be monitored. How this is done is left to implementation (as an example, the PMS could be part of the IGP domain or it could monitor the IGP by BGP-Link State [RFC9085]).

Knowing the IGP topology enables ~~the-a~~ PMS to steer measurement packets along arbitrary pre-determined concatenated sub-paths, identified by suitable Segment IDs (SIDs). The SR connectivity metric specified ~~below in this document~~ requires setting up a number of constrained, overlaid measurement loops (or measurement paths). The delay of the packets sent along each of these measurement loops is captured. A single congested interface along a monitored sub-path adds latency to a unique subset of several measurement loops. If a monitored sub-path no longer provides connectivity between two nodes, a unique subset of measurement loops will indicate drop of all traffic while connectivity is lost. The number of measurement loops required in total may be limited to one per sub-path (or connection) to be monitored, if a hub-and-spoke like sub-path topology as described below is monitored. In addition to information revealed by a commodity ICMP ping measurement, the metrics and methods specified here identify the location of a congested interface (or sub-path, respectively).

The measurement loop packets remain in the data plane of ~~passed~~involved routers. Hence, routers only forward ~~the~~ measurement packets without additional processing.

It is recommended to consider automated measurement loop set-up. The methods ~~proposed~~defined here are error-prone, if the topology and measurement loop design isn't applied properly. While details of an automated set-up are not within scope of this document, some formal ~~defintions~~definition of constraints to be respected are given.

This document specifies ~~T~~type-p metrics determining properties of an SR path which allows to monitor connectivity and congestion of network element interfaces. The specified methods further allow to locate the path or interface which caused an anomaly in the reported ~~type~~Type-p metrics.

This document is limited to the Segment Routing MPLS layer, but the methodology may be applied within SR domains or MPLS domains in general.

1.1. Requirements Language

Commenté [MB3]: Move to an Operational Considerations Section.

Commenté [MB4]: Add a ref

The key words "MUST", "MUST NOT", "REQUIRED", "SHALL", "SHALL NOT", "SHOULD", "SHOULD NOT", "RECOMMENDED", "NOT RECOMMENDED", "MAY", and "OPTIONAL" in this document are to be interpreted as described in BCP 14 [RFC2119] [RFC8174] when, and only when, they appear in all capitals, as shown here.

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2. Experiment Goals and Success Criteria

A goal of the experiment is to strengthen the operational and deployment recommendations gathered in both controlled domains and real operational conditions. Such experimental work is also needed to assess the required integration effort for the support of the new metric.

Parties interested in executing the experimental monitoring approach described by this document should publish the results of their experiments. Set-up and publication of results should document the following:

- * Set up an SR lab network (or use an operational network).
~~Indueing~~Inducing the events to be detected by the suggested monitoring metric is simpler in a laboratory.
- * Deploy or develop a ~~Path Monitoring System (PMS, -) see [RFC8403]~~.
Note, ~~that~~ no changes in existing ~~Segment Routing~~SR protocols, standards are required to ~~program~~program a PMS and connect it to an SR domain.
- * Validating the SR domain monitoring approach defined below requires SR domain IGP / control plane topology awareness. The PMS needs to allow for ~~configuartion~~configuration of the measurement loop SR segment stacks. Manual monitoring loop set-up is sufficient to prove that the approach defined by this document works.
- * Nodes need to forward packets with an SR segment stack depth of 6 segments (or 5 as minimum, if the PMS is directly connected to two "hub-nodes").
- * The experiment requires standard SR routing and forwarding. Activation of Adj-SID announcement in IGPs may be ~~usful~~useful and simplify setting up the PMS monitoring loops.
- * Apart from requesting ~~the-a~~ PMS to create standard-conformant SR packets, there are no interoperability requirements.
- * The measurement loop design ~~traget~~targets ~~en-~~creating a limited amount of measurement traffic. ~~Sealeability~~Scalability constraints may arise from the performance constraints of ~~the-a~~ PMS. Monitored nodes and interfaces are expected to support standard SR packet forwarding.

* ~~Effectiveness~~Effectiveness of the ~~proposed~~-SR connectivity monitoring metric, including:

- Could the ~~Round-Round-Trip~~ Delay be determined for each monitored link as defined below?
- Could the single loss of a monitored link be detected as defined below?
- Could ~~sengestion~~congestion of a single monitored node-interface be detected as defined below?

3. A ~~segment~~Segment ~~routing~~Routing ~~connectivity~~Connectivity ~~monitoring~~Monitoring ~~framework~~Framework

The ~~Segment-Routing~~SR IGP topology information consists of the IP and (if present) the MPLS layer topology. The minimum SR topology information consists of Node-Segment-Identifiers (Node-SID), identifying SR routers. The IGP exchange of Adjacency-SIDs (Adj-SID) [RFC8667], which identify local interfaces to adjacent nodes, is optional. It is RECOMMENDED to distribute Adj-SIDs in a domain operating a PMS to monitor connectivity as specified below. If Adj-SIDs aren't ~~availbale~~available, [RFC8287] provides methods how to steer

packets along desired paths by the proper choice of an MPLS Echo-request IP-destination address. A detailed description of [RFC8287] methods as a replacement of Adj-SIDs is out of scope of this document. Monitoring interfaces connecting nodes requires Adj-SIDs, if re-converged IP/MPLS layer connectivity would result in re-routing packets (and re-establishment of IP/MPLS layer connectivity) by using Node-SIDs.

An active ~~round-round-trip~~ measurement between two adjacent nodes is a simple method to monitor connectivity of a connecting link. If multiple links are operational between two adjacent nodes and only a single one ~~loses~~loses connectivity, a single plain ~~round-trip~~round-trip measurement may fail to notice that or fail to identify which link has lost connectivity. A ~~round-trip~~round-trip measurement further fails to identify which ~~particular~~-interface is congested, even if only a single link connects two adjacent nodes.

~~Segment-Routing~~SR enables the set-up of extended measurement loops. Several different measurement loops can be set up to form a partial overlay. If done ~~properly~~adequately, any network change impacts more than a single measurement loop's ~~round-round-trip~~ delay or causes drops of packets

of more than one loop. Randomly chosen measurement loop paths including the interfaces or paths to be monitored may fail to produce the desired unique result patterns, hence commodity network tomography methods aren't applicable [CommodityTomography]. The approach pursued here uses a pre-specified measurement loop overlay design to produce the desired results with a limited number of measurement loops.

A centralised monitoring approach doesn't require report collection and result correlation from two (or more) receivers. The metrics captured along different measurement loops however need to be correlated.

A further characteristic of the measurement loop configuration defined below is that it enables estimating the packet round trip delay for a monitored link or sub-path.

An example hub and spoke network, operated as SR domain, is shown [below in Figure 1](#). The PMS attached to L300 is supposed to monitor the connectivity of the 6 links attaching the spoke-nodes L050, L060 and L070 to the hub-nodes L100 and L200. L300 only serves to connect the PMS to nodes L100 and L200. A link may be taken as a simple and generic kind of sub-path.

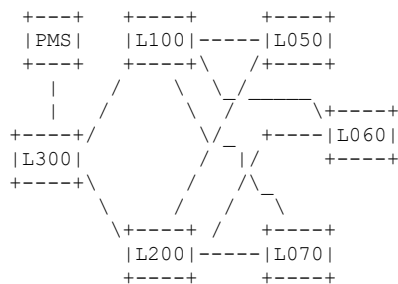


Figure 1: [Sample Topology](#)

Example hub and spoke network allowing link connectivity verification with a PMS

The SID values are picked for convenient reading only. Node-SID: 100 identifies L100, Node-SID: 300 identifies L300 and so on. Add [definitions](#) for Adj-SID 10050: Adjacency L100 to L050,

Adj-SID

10060: Adjacency L100 to L060, Adj-SID 60200: Adjacency L60 to L200 and so on (note that the Adj-SID are locally assigned per node interface, meaning two per link).

Monitoring the 6 links between hub nodes Ln00 (where n=1,2) and spoke nodes L0m0 (where m=5,6,7) requires 6 measurement loops, which have the following properties:

- * Each measurement loop passes a single round trip from one hub Ln00 to one spoke L0m0 (e.g., from L100 to L050 and back from L050 to L100).
- * Each measurement loop passes two more links: one from the same hub Ln00 to another spoke L0m0 and from that spoke to the alternate hub Lm00 (e.g., from L100 to L060 and then from L060 to L200)
- * Every monitored link is passed by a single round trip measurement

loop only once and further only once unidirectional by two other loops. These two unidirectional measurement loop sections forward packets in opposing direction along the monitored link. In the end, three measurement loops pass each single monitored link (or sub-path, respectively). In figure 1, e.g. the link between L100 and L050 is passed by one measurement loop following a round trip L100 to L050 (the measured delay is M1, see below), a second loop passes in direction L100 to L050 only (delay M3) and a third loop passes in direction L050 to L100 only (delay M6). See below for the measurement loops M1 - M6.

Note that any 6 links connecting two to five nodes can be monitored that way too. Further note that the measurement loop overlay chosen is optimised for 6 links and a hub and spoke topology of two to five nodes. The 'one measurement loop per measured sub-path' paradigm only works for the given topology.

The above scheme results in 6 partially overlaying measurement loops for the given example. The start and end of each measurement loop is PMS to L300. ~~Paths~~Paths continue to L100 or L200 and finally return by a similar sub-path in the opposite direction. These parts of the measurement loops are omitted here for brevity (some discussion may ~~be found~~be found below). The following delays are measured along the SR paths of each measurement loop:

1. M1 is the delay along L100 -> L050 -> L100 -> L060 -> L200
2. M2 is the delay along L100 -> L060 -> L100 -> L070 -> L200
3. M3 is the delay along L100 -> L070 -> L100 -> L050 -> L200
4. M4 is the delay along L200 -> L050 -> L200 -> L060 -> L100
5. M5 is the delay along L200 -> L060 -> L200 -> L070 -> L100
6. M6 is the delay along L200 -> L070 -> L200 -> L050 -> L100

For the sake of simplicity, in the following delay M1 also identifies the corresponding measurement loop number 1 and so on.

An example for a stack of Adj-SID segments the loop resulting in M1 is (top to bottom): 100 | 10050 | 50100 | 10060 | 60200 | PMS. As can be seen, the Node-SIDs 100 and PMS are present at top and bottom of the segment stack. Their purpose is to transport the packet from the PMS to the start of the measurement loop at L100 and return it to the PMS from its end. When connectivity is lost, a path determined by Adj-SIDs behaves deterministic: packets forwarded to an Adj-SID without connectivity to the neighboring node are dropped.

An example for a stack of a loop consisting of Node-SID segments allowing to capture M1 is (top to bottom): 100 | 050 | 100 | 060 |

Evaluation of the measurement loop round trip delays M1.. to M6 allows to detect the ~~following~~following state-changes of the monitored sub-paths:

- * If the loops are set up using Node-SIDs only, any single complete loss of connectivity caused by a failing single link between any Ln00 and any L0m0 node briefly disturbs three measurement loops and changes the delay measured along them. The traffic to the Node-SIDs is re-routed (in the case of a single link loss, no node is completely disconnected in the example network). In this mode of operation, a suitable metric ~~characterising~~characterizing re-routing coupled with the loss of that single link is required. The change in propagation delay might be an approach for such a metric (if there is any delay change - that depends on the delay of the resulting alternate route). A delay based ~~connectivity~~connectivity scheme may not work under all circumstances.

- * If the measurement loops are set up using Adj-SIDs only, a loss of connectivity caused by a failing single link between any Ln00 and any L0m0 node terminates the traffic along three measurement loops. The packets of all three loops will be dropped, until the link gets back into service. Traffic to Adj-SIDs is not rerouted. As mentioned above, Node-SIDs may be used to ~~feward~~forward the measurement packets from the PMS to the hub node, where the first measurement loop containing links or sub-paths to be monitored begins and from the hub node receiving a measurement packet from the last monitored sub-path segment to the PMS.

- * This simple example indicates superiority of Adj-SIDs over Node-SIDs only if links are monitored and the network architecture is ~~similiar~~similar to the one shown in the figure. The generic advice is to base unambiguous connectivity monitoring preferably on packet loss evaluation (using Adj-SIDs to address the monitored links). Distinct delay changes may result as a consequence of re-route (if Node-SIDs were applied to address the monitored links). The resulting delay changes needed to be pre-calculated to ensure that this condition holds.

- * A single congested interface (or otherwise hampered forwarding) between any Ln00 and any L0m0 node always only impacts the measured delay of two measurement loops.

- * As an example, the formula to calculate the (sub-path) Round Trip Delay (RTD) for link L100-L050 is given here:

$$4 * RTD_{L100-L050-L100} = 3 * M1 + M3 + M6 - M2 - M4 - M5.$$

This formula is reproducible for all other links: sum up 3*RTD measured along the loop passing the monitored link of interest in round trip fashion, and add the RTDs of the two measurement loops passing the link whose RTD is calculated only in a single direction. From this sum subtract the RTDs captured for the

measurement loops not passing the monitored link under RTD evaluation. The result is four times the RTD of the link evaluated for its delay.

- * If blackhole routes were to be detected, the sub-paths to be monitored need to be set-up using Node-SIDs. In a strict sense, i.e., detecting blackhole routes only, single losses of physical connectivity along any sub-path and delay changes caused by it may be ignored. If just a single of the monitored routes is blackholed by one router, the change to be detected will impact two measurement loops. If any one router generally causes blackhole routing, the resulting pattern depends on its role: a hub-router will disturb all measurement loops, while a spoke disturbs only four measurement loops.
- * Scenarios where setting up monitoring loops with a mix of Adj- and Node-SIDs are left for further study.

Any single event of interest for the proposed metric, which are a single loss of connectivity or a single case of congestion, only impacts a unique set of measurement loops which can be determined a-priori. If, e.g., connectivity is lost between L200 and L050, measurement loops M3, M4 and M6 indicate packet loss (or a change of the measured delay, if a Node-SID based approach is preferred).

As a second example: if the interface L070 to L100 is congested, measurement loops M3 and M5 indicate a change in the measured delay. Without listing all events, it can be shown that all cases of single losses of connectivity or single events of congestion influence only delay measurements of a unique set of measurement loops.

The measurement loops are best set up while there's no congestion. In the absence of congestion, the RTDs of all monitored links can be calculated as shown above. Doing so may allow queue-depth estimates, once congestion occurs. A single congestion event adds queuing delay to the RTD measured of two specific measurement loops. The two measurement loops impacted indicate the congested interface and enable estimation of the queue-depth (in terms of seconds based on comparing actual and prior delay measurements). Assume the per link RTD to have been calculated while the network was not congested at interval T0. As an example, a queue of an average depth of 20 ms may build up at interface L200 to L070 at a later interval T1. The measurement loops M5 and M6 are the only ones passing the interface in that direction. Both indicate an added delay along M5 and M6 of +20 ms during this measurement interval T1 with congestion on this interface, while M1-4 indicate unchanged delays. The location of the congested interface is determined by the combination of the two (and only two) measurement loops M5 and M6 showing a significant delay increase. The average queue depth [s] = $(M5[T1] - M5[T0] + M6[T1] - M6[T0]) / 2$.

As mentioned above, there's a constant delay added for each measurement loop, which is the delay of the path passed from PMS -> L100 + L200 -> PMS. Please note, that this added delay is appearing twice in the formula resulting in the monitored link delay estimate of the example network. Then it is the RTD PMS -> L100 + RTD L200 -> PMS. Both RTDs can be directly measured by two additional measurements Cor1 = RTD (PMS -> L100 -> PMS) and Cor2 = RTD (PMS ->

L200 -> PMS). The monitored link RTD formula was $\text{linkRTDuncor} = 3 \cdot M_x + M_y + M_z - M_s - M_t - M_u$. The correct $4 \cdot \text{linkRTD}_x = 4 \cdot \text{linkRTD}_{x\text{uncor}} - \text{Cor1} - \text{Cor2}$.

If the interface between PMS and L100/L200 is congested, all measurement loops M1-M6 as well as Cor1 and Cor2 will see a change. A congested interface of a monitored link doesn't impact the RTDs captured by Cor1 and Cor2.

The measurement loops may also be set up directly connecting the PMS to the hub nodes L100 and L200, if that's preferred and supported by the network architecture. In that case, the above formulas apply without correction.

4. Topology and measurement loop set up requirements

4.1. General network topology requirements

The metric and methods specified below can be applied to monitor networks or sub-paths forming a hub and spoke topology. A single sub-path status change of type loss of connectivity or congestion can be detected. The nodes don't have to act as hubs or spokes, this terminology is only chosen to describe a topology requirement. In detail, the topology to be monitored MUST meet the following constraints:

- * The SR domain sub-paths to be monitored create a hub and spoke topology with a PMS connected to all hub nodes. The PMS may reside in a hub.
- * Exactly 6 (six) sub-paths are monitored.
- * The monitored sub-paths connect at least two and no more than 5 nodes.
- * Every spoke node MUST have at least one path to every hub node.
- * Every spoke node MUST at least be connected to one (or more) hub node(s) by two monitored sub-paths.
- * Sub-paths between spokes can't be monitored and therefore are out of scope (the overlay measurement loops can't be set up as desired).

Shared resources, like a Shared Risk Link Group (e.g., a single fiber bundle) or a shared queue passed by several logical links, need to be considered during set up. Shared resources may either be desired or to be avoided. As an example, if a set of logical links share one parental scheduler queue, it is sufficient to monitor a single logical connection to capture the state of that parental scheduler.

4.2. Sub-path Monitoring measurement loop routing requirements

The methodologies ~~sepeified~~specified by this document REQUIRE a measurement loop path overlay of all path delay measurement streams F_i , i in $[1, 2 \dots 6]$ as defined in this section. In the ~~following~~following, a path delay

measurement stream Fi is called measurement (loop) Fi for brevity.

- * Define the segment routed Sub-paths SPi, i in [1, 2...6] to be monitored. The Sub-paths SPi SHOULD not share resources, if the operator isn't aware of the impact of the shared resources on the measurement loops Fi and the methodologies defined below. The Sub-path SPi topology SHOULD respect the general network topology requirements as specified above.
- * Set up i = 1, 2...6 measurement loops Fi thus that measurement Fi passes SPi and only SPi bidirectional (or by a round-trip) from Hub to Spoke and back. Note that the ~~correspondance~~correspondence of SPi and Fi isn't strictly required. By this ~~correspondance~~correspondence, measurement Fi however appears in all methodologies calculating a metric related to SPi.
- * Set up the SR path per measurement loops Fj and Fk so that SPi is passed by exactly one other measurement loop Fj unidirectional in direction Hub to Spoke and by exactly one other measurement loop Fk unidirectional in the opposite direction (Spoke to Hub). The measurement loop Fi != Fj != Fk. As a description, one measurement loop Fj pass SPi in "downstream" direction from Hub to Spoke, whereas measurement loop Fk passes SPi in "upstream" direction from Spoke to Hub.
- * Set up each segment routed measurement loop path Fi thus that it passes SPi bidirectional as specified above, SPj unidirectional from Hub to Spoke and SPk unidirectional from Spoke to Hub. The monitored Sub-path SPi MUST NOT be equal to SPj and MUST NOT be equal to SPk.
- * The measurement loop set up to monitor all Sub-paths SPi is completed, if:
 - + Each Sub-path SPi is passed by exactly three measurements loops Fi, Fj and Fk as specified above.
 - + Each segment routed measurement loop path Fi passes exactly three concatenated Sub-paths SPi, SPm and SPn as specified above (indices m and n are chosen here only to avoid misconceptions which may result from picking indices j and k already appearing before - equality of j and k with either m and n is neither excluded nor required).

4.3. Path

This document specifies sub-path monitoring within a closed domain by a controlled and pre-designed measurement loop set-up. The path traversed by the packet SHOULD be detected. Verifying that the forwarding path is in line with the desired measurement loop set-up ensures accurate evaluation of the metric. See [RFC8287] for SR MPLS OAM and [RFC9259] for SRv6 OAM.

4.4. Sub-path Monitoring measurement loop packet spacing

Packets per measurement loop Fi are sent periodically by a temporal

distance of IncT. For convenience, packets of the 6 measurement loops are assumed to be equally spaced at the sender too. Let's define the temporal distance IncF between two consecutive packets sent along to different measurement loops Fi and Fj at a single sender to be

$$\text{IncF} = \text{IncT} / 6$$

Further it seems useful to suggest IncF to be bigger than the largest measurement loop delay max (mi) under stable network operation (i.e., including some tolerance). Further assume the standard deviation of the measurement values mi to be much smaller than the delay mi, which is likely for a sub path being a regional or national link in many countries. Note that this definition isn't a strict requirement. Interpretation of results is however simplified by it. For the rest of the document assume

$$\text{IncF} > 2 * \max (mi), i \text{ in } [1...6], \text{ which results in}$$

$$\text{IncT} > 12 * \max (mi)$$

Discussion and reasoning for a reasonable smallest interval IncF in relation to max(mi) follows below.

5. Generic Type-P-SR-Path-Periodic-* metric

To reduce the redundant information presented in the detailed metrics sections which follow, this section presents the specifications shared by two or more metrics. To simplify comparisons, this section is structured using the same subsections as the individual metrics.

5.1. Metric Name

All metrics use the Type-P convention as described in [RFC2330]. The rest of the name is unique to each metric.

5.2. Generic Metric Parameters

Refer to section 3.2. Metric Parameters: Type-P-* of [RFC6673]. The following parameters are added, enhanced or removed:

Dst SHOULD be a diagnostic IP address as specified by [RFC8287] and [RFC8029], if MPLS OAM is operated to capture the metric.

Fi, where i in [1, 2...6], a selection function that unambiguously defines a packet of one particular stream i, which forms part of the monitoring overlay measurement loop set up.

L, a packet length in bits. All packets of Type-P-SR-Path-Delay-Periodic-Streams Fi SHOULD be of the same size.

MLAi, a stack of Segment IDs determining a monitoring loop Fi. The Segment-IDs MUST be chosen so that a singleton type-p packet of selection function Fi passes the sub-path i to be monitored.

No support: lambda (Poisson Streams remain ffs.)

5.3. Metric Units

Refer to section 3.4. Metric Units: Type-P-* of [RFC6673].

6. Singleton Definition for Type-P-SR-Path- Periodic-Delay

6.1. Metric Name

Type-P-SR-Path-Periodic-Delay

6.2. Metric Parameters

See ~~section~~ Section 5.2.

6.3. Delay Metric Units

A sequence of consecutive time values. The value of a Type-P-SR-Path-Periodic-Delay is either a real number or an undefined (informally, infinite) number of seconds per singleton of each stream Fi.

6.4. Definition

Section 3.4 of [RFC7679] applies per singleton of each stream Fi. The additional information related to singletons of section 4.2.4 of [RFC3432] applies too.

6.5. Discussion

See section 3.5 of [RFC7679]. One generalisation seems appropriate: a global satellite navigation system affords one way to achieve synchronization within usec.

6.6. Methodologies

Section 3.6 of [RFC7679] applies per stream Fi with one exception: at the Src host, select Src and Dst IP addresses, if IP-routing is applied, or select the proper functional IP-destination address if an [RFC8287] SR MPLS OAM packet format is applied. Further add the appropriate stack of Segment IDs MLAi determining the monitoring loop Fi and form a test packet of Type-P with these addresses and the segment stack.

6.7. Errors and Uncertainties

See Section 3.7 of [RFC7679] and ~~section-Section~~ 4.6 of [RFC3432].

6.8. Reporting the metric

See ~~section-Section~~ 3.8 of [RFC7679].

7. Singleton Definition for Type-P-SR-Path- Packet-Loss

Editors note: To be added based on existing loss metrics. A delay based approach indicating loss of a physical interface by detecting delay changes caused by re-routing can't be assumed to reliably cause unique delay change patterns under all circumstances (consider a shortest path routed multi-hop MPLS sub-path to be monitored rather than a link or a scenario where a bundle of 6 equivalent links is

monitored connecting a single hub and spoke).

7.1. Metric Name

Type-P-SR-Path-Packet-Loss

7.2. Metric Parameters

See ~~section~~ Section 5.2.

7.3. Packet Loss Metric Units

The value of a Type-P-SR-Path-Packet-Loss is either a zero (signifying successful transmission of the packet) or a one (signifying loss) per singleton of each stream Fi.

7.4. Definition

Section 2.4 of [RFC7680] applies per singleton of each stream Fi.

7.5. Discussion

See ~~section~~ Section 3.5 of [RFC7680].

7.6. Methodologies

Section 2.6 of [RFC7680] applies per stream Fi with one exception: at the Src host, select Src and Dst IP addresses, if IP-routing is applied, or select the proper functional IP-destination address if an [RFC8287] SR MPLS OAM packet format is applied. Further add the appropriate stack of Segment IDs ML*A*i determining the monitoring loop Fi and form a test packet of Type-P with these addresses and the segment stack.

7.7. Errors and Uncertainties

See section 2.7 of [RFC7680].

7.8. Reporting the metric

See section 2.8 of [RFC7680].

8. Definition of Samples for Type-P-SR-Path-Periodic-Delay

This sections defines metric samples and metrics derived from samples.

8.1. Generic Type-P-SR-Path-Periodic-Delay-* metric

To reduce the redundant information presented in the detailed metrics sections that follow, this section presents the specifications that are common to two or more metrics. The section is organized using the same subsections as the individual metrics, to simplify comparisons.

8.1.1. Metric Name

Type-P-SR-Path-Periodic-Delay-*

8.1.2. Metric Parameters

Src, the IP address of a host

Dst, the IP address of a host

MLAi, a stack of Segment IDs

Ti0, a time

Tif, a time

incT, a time

8.1.3. Metric Units

See section Section 6.3.

8.1.4. Metric Defintion

Given Ti0 and Tif and nominal inter-packet interval incT, those time values greater than or equal to Ti0 and less than or equal to Tif are then selected. At each of the selected times in this process, we obtain one value of Type-P-SR-Path-Periodic-Delay. The value of the sample is the sequence made up of the resulting [time, delay] pairs. If there are no such pairs, the sequence is of length zero and the sample is said to be empty.

8.1.5. Discussion

See section 4.4 of [RFC3432].

8.1.6. Errors and uncertainties

See section 4.6 of [RFC3432].

8.2. Definition of Type-P-SR-Path-Periodic-Delay-Stream

The only definition required for this metric is a unique metric name.

8.2.1. Metric Name

Type-P-SR-Path-Periodic-Delay-Stream

8.3. Definition of Type-P-SR-Path-Periodic-Delay-Variation

The smallest sample Type-P-SR-Path-Periodic-Delay-Stream is one of two consecutively received values. These may be used to calculate a Segment Routed Path Delay-Variation (SRDV) singleton, defined below.

8.3.1. Metric Name

Type-P-SR-Path-Periodic-Delay-Variation

8.3.2. Methodologies

SRDV[i,j], for each sample of packets j and j-1 of stream Fi, j > 1,

the delay variation between successive packets is calculated as:

$$\text{SRDV}[i,j] = \text{Delay}[i,j] - \text{Delay}[i,j-1],$$

j in $[2,3\dots N]$ and N the total number of packets received at Dst. If one or more of the M packets sent by Src are lost, they are ignored for the metric, as no reasonable metric value is defined here. If $N > 1$, the metric is calculated for every valid packet received and the preceding one.

8.3.3. Discussion of SRDV

Evaluation statistics of differential SRDV metric samples may help to identify issues.

8.3.4. Errors and uncertainties

See section 2.7 of [RFC3393].

8.4. Definition of Type-P-SR-Path-Periodic-Delay-Variation-Stream

The only definition required for this metric is a unique metric name.

8.4.1. Metric Name

Type-P-SR-Path-Periodic-Delay-Variation-Stream

8.4.2. Metric ~~Defintion~~Definition

Given $Ti0$ and Tif , those time values greater than or equal to $Ti0$ and less than or equal to Tif are then selected. At each of the selected times in this process, we obtain one value of Type-P-SR-Path-Periodic-Delay. The value of the sample is the sequence made up of the resulting [time, delay-variation] pairs with time being set to the Dst timestamp of the Delay-Variation singleton, for which a valid singleton is calculated. If there are no such pairs, the sequence is of length zero and the sample is said to be empty. If N Delay singletons are captured and sampled $N-1$ Delay-Variation singletons are sampled during the same interval

9. Statistic Definitions for SR-Path-Periodic-*-Stream samples

Change point detection requires statistical ~~defintions~~definitions. These are provided below. The names of the statistics contain an "*" placeholder, which may be replaced by "Delay" or "Delay-Variation".

9.1. SR-Path-Periodic-*-Mean

For a type-p metric, the mean is specified by:

$$\text{SR-*Mean} = (1/N) * \text{Sum}(\text{from } a=1 \text{ to } N, \text{value}[a])$$

* N sample size

* value sample value of a sampled [time, value] pair

9.2. SR-Path-Periodic-*-Std

For a type-p metric, the Standard-Deviation Std is specified by:

$$SR\text{-}Std = [1/(N-1)] * \text{Sum}(\text{from } a=1 \text{ to } N, [SR\text{-}Mean - \text{value}[a]]^2)$$

- * N sample size
- * value sample value of a sampled [time, value] pair
- * SR-Mean sample mean of the same metric as defined above

The definition as given above requires a two-pass calculation per sample. Algorithms estimating the standard-deviation by one-pass calculation have been published and might be preferable, if metric singletons and samples aren't buffered or calculations need to be fast.

10. Statistic Definitions for Type-P-SR-Path-Packet-Loss

The packet loss ratio is a useful metric to ~~characterise~~characterize congestion.

10.1. SR-Path-Packet-Loss-Ratio

See ~~section~~Section 4.1 of [RFC7680]

11. Sub-Path monitoring metrics derived from samples captured along the measurement loops

To produce meaningful sub-path monitoring values, the measurement loop metrics are captured during a phase with stable networking conditions. In a backbone network domain, the absence of congestion often is a sufficient condition (frequent traffic shifts due to changes in routing and traffic engineering aren't expected). This may be different in a network based on a shared medium. It may be outright difficult in networks with frequently changing traffic management- and routing-policies.

In the following, the index CS indicates a statistic captured during a ~~mesurement~~measurement interval with stable routing and no congestion.

11.1. Baseline measurement

Capture a sample of delay values Type-P-SR-Path-Periodic-Delay-Stream of sample size N for each ~~measurment~~measurement loop Fi. As a rule of thumb choose N in [30, 100].

For each measurement loop Fi, calculate the following metrics ~~characterising~~characterizing the monitored Sub-Paths during stable and congestion free network conditions:

- * SR-Path-Delay-MeanCSi, the mean delay captured along measurement loop Fi

- * SR-Path-Delay-StdCSi, the standard-deviation of the delay captured along measurement loop Fi
- * SR-Path-Delay-Variation-MeanCSi, the mean delay variation captured along measurement loop Fi
- * SR-Path-Delay-Variation-StdCSi, the standard-deviation of the delay variation captured along measurement loop Fi

A stable and uncongested network should produce rather constant delays, resulting in low standard-deviation values and almost zero mean delay variation. [Editors note: Add text to select the median of a small set of stream mean captures, like 5 samples captured consecutively.]

Example data was captured in a lightly loaded Gigabit network. 11 routers are passed per measurement loop. The sample size is 30 packets, more than 200 samples were captured per measurement loop. The loops are set up for a different purpose than specified here, they are picked due to a high number of passed routers. Note that SR-DV-Mean here refers to an abs(SR-DV-Mean) sample, thus small, positive, non-zero means result. The time unit is microseconds.

Metric	Quantile	SR-D-Mean	SR-D-Std	SR-DV-Mean	SR-DV-Std
Loop1	95%	34507	62	41	84
Loop2	95%	35104	45	34	49
Loop1	50%	34496	19	19	17
Loop2	50%	35088	15	14	12
Loop1	5%	34491	14	20	12
Loop2	5%	35080	13	12	9

Figure 2

Example baseline metrics for an 11 hop measurement loop (quantiles refer to SR-D-Mean)

11.2. Discussion of the baseline measurement

Delay outliers may occur at any time in any communication network, and the measurement system packet processing itself may also produce some. It is fair to expect only single outliers in a stable, not congested network. It may be worth to capture several consecutive SR-Path-Periodic-*--Stream samples and compare their statistics, before picking reasonable baseline metric values. Samples showing higher standard deviations (compare the 95% quantile values in the above figure to the 50% quantile values) may benefit from removing the maximum singleton value from the sample. This will smooth the mean and standard-deviation, and if the result then is closer to those of the majority of the samples, foster confidence in

determining the baseline metrics. Depending on the preferred method of data-processing and storing, this may require capturing the sample maximum as a separate metric.

11.3. Definition of SR-Path-Sub-Path-RTD-Estimate

Within a single evaluation interval of identical Time T_0 and T_f , SR-Path-Delay-MeanCSI (from now on DMeanCSI) is the mean delay of the measurement loop passing the monitored Sub-Path S_{Pi} by a round trip. Let's keep the indexing applied above, then F_j and F_k with captured mean delays DMeanCSj and DMeanCSk pass S_{Pi} unidirectional. Further, 3 measurement loops F_x , F_y and F_z don't pass Sub-Path S_{Pi} at all. The corresponding mean delays are DMeanCSs, DMeanCSt and DMeanCSu.

The the SR-Path-Sub-Path-RTD-Estimate of the ~~Round-Trip~~Round-Trip Delay along the monitored Sub-Path F_i , RTD_{Fi} , is

$$RTD_{Fi} = (3 * DMeanCSI + DMeanCSj + DMeanCSk - DMeanCSx - DMeanCSy - DMeanCSz) / 4$$

11.4. Definition of SR-Path-Sub-Path-* -Changepoint

The asterisk stands for "Interface" as well as "Connectivity". If connectivity is lost and no path is available between two nodes, any packets to be transmitted will be dropped. A change in sub-path routes with a change in measurement loop delay ~~indicates~~indicates a re-~~routing~~routing event (a temporal loss in connectivity), not a long lasting loss of connectivity. Hence a change in measurement loop delays caused by a re-routed monitored sub isn't useful to derive a metric indicating connectivity loss on a monitored sub path (a sub-path-route-change metric might be of interest, but isn't within scope of this document).

Network changes like congestion or re-routing are often ~~characterised~~characterized by a change in the mean delay of a monitoring measurement. CUSUM (cumulative sum~~→~~) charts have been shown to be efficient in detecting shifts in the mean of a process [NIST]. The upper bound CUSUM is defined as:

$$Sup(t) - Fi - Delay = \max(0, Sup(t-1) + x_t - SR-Path-* - MeanCSI - k_i)$$

with $Sup(0) = 0$, $k_i = \Delta * SR-Path-* - StdCSI$ (Δ is a dimensionless integer number), $x_t = Type-P-SR-Path-Periodic-*$ singleton for measurement loop F_i at time t .

The actual SR-Path-Delay-Mean of Measurement Loop F_i is decided to be significantly above $SR-Path-* - MeanCSI$, if:

$$Sup(t) - Fi - Delay > h_{SP}, \text{ with } h_{SP} = d * k_i \text{ (d is a dimensionless integer number).}$$

An **analogous CUSUM** controls changes to a lower mean delay (which may be caused by a re-routing event):

$$Slo(t) - Fi - Delay = \max(0, Slo(t-1) + SR-Path-* - MeanCSI - x_j - k)$$

a mis en forme : Surlignage

The actual SR-Path-Delay-Mean of F_i is decided to be significantly below SR-Path- $\ast$ -MeanCSI, if:

$Slo(t) - F_i - Delay > h_{SP}$

11.5. Discussion of SR-Path-Sub-Path- $\ast$ -Changepoint

CUSUM chart based changepoint detection is sensible even to small changes in the mean. CUSUM charts offer a limited protection against single, isolated outliers. A cumulated sum only grows, if the ~~controled~~controlled process ~~consistenly~~consistently changes its mean (or standard deviation, respectively). Assuming constant physical minimum delays to ~~characterise~~characterize wireline communication networks, a change in standard deviation not affecting the mean delay doesn't seem to be caused by a change in networking conditions.

The measured delays will change once a Sub-Path route has changed, or once persistent congestion starts to fill a queue. Both indicate changes in the network. As the Sub-Pathes SPI form an overlay with designed properties, every network change affecting a sub-path creates correlated SR-Path- $\ast$ metric changes. As the ~~correspondance~~correspondence of network changes to Sub-Path metrics is known a-priory, detecting correlated SR-Path- $\ast$ metric changes allows to locate the change.

In the absence of packet re-routing, packet loss is ~~characterising~~characterizing a loss of connectivity. Packet loss requires a time threshold when to decide that an active measurement packet was lost, and consecutive loss requires receiver awareness, that packets have been sent (this argues for the sender to be the receiver, unless both ~~communicate~~communicate fast and reliable out of band).

The preferred CUSUM ~~parametrisation~~parametrization will depend on the kind of events to detected and on the outlier characteristics.

$k_i = \Delta * SR-Path- \ast - StdCSI$ may be set to a value relevant high enough to exclude single outliers to trigger an alert, but low enough to indicate persistent changes in delay. The same holds for the to be picked for d .

A broader discussion on CUSUM parametrization may be found in literature. Networking skills are required to parametrise CUSUM, as well as to ~~interprete~~interpret the results (notably to differ re-routing from congestion).

11.6. Definition of SR-Path-Sub-Path-Congestion-Location

An interface along a single monitored Sub-Path SPI whose queue is persistently filled adds latency to measurement loop F_i and one of the two unidirectional measurement loops F_j and F_k passing Sub-Path SPI . F_j has been defined to pass SPI from Hub to Spoke and F_k pass SPI in opposite direction. Then SR-Path-Sub-Path-Congestion-Location

metric for the traffic directed from "Hub to Spoke" along Sub-Path SPi is:

$$SPi_ConLoc_ij = Sup(t)_SPi_Periodic-Delay + Sup(t)_SPj_Periodic-Delay$$

And for the opposite traffic direction, from "Spoke to Hub":

$$SPi_ConLoc_ik = Sup(t)_SPi_Periodic-Delay + Sup(t)_SPk_Periodic-Delay$$

Note that another 10 SR-Path-Sub-Path-Congestion-Location metrics are calculated, one per monitored Sub Path and traffic direction. The evaluation can be simplified as follows:

```
IF SPi_ConLoc_ij > h_SP
AND h_SP > Sup(t)_SPk_Periodic-Delay
AND h_SP > Sup(t)_SPx_Periodic-Delay
AND h_SP > Sup(t)_SPy_Periodic-Delay
AND h_SP > Sup(t)_SPz_Periodic-Delay
```

Then Sub-Path SPi faces congestion in direction "Hub to Spoke".

```
IF SPi_ConLoc_ik > h_SP
AND h_SP > Sup(t)_SPj_Periodic-Delay
AND h_SP > Sup(t)_SPx_Periodic-Delay
AND h_SP > Sup(t)_SPy_Periodic-Delay
AND h_SP > Sup(t)_SPz_Periodic-Delay
```

Then Sub-Path SPi faces congestion in direction "Spoke to Hub".

Here, h_SP is a universal threshold in unit time to indicate a filling queue or a significant change in delay due to a Sub-Path reroute or another persistent change in topology (like e.g. automated Layer 1 / Layer 2 topology changes). Packets following SPx, SPy and SPz don't pass the congested interface of Sub-Path SPi.

11.7. Definition of SR-Path-Sub-Path-Disconnected

The idea of this document is to monitor a set of sub-paths for a single case of congestion or a single loss of connectivity. If a single sub-path SPi ~~loses~~loses connectivity, i.e., all packets are dropped in both sub-path forwarding directions, then three measurement loops m_i , m_j and m_k fail to receive any traffic. A single interface congestion will add latency to m_i and one of m_j or m_k , respectively. Still, if it is congestion of a single sub-path SPi interface causing additional latency, either m_j or m_k face no congestion and the one measured delay m_j or m_k should be within the expected range of values. Rather than basing a loss of connectivity metric on a "reliable" indication SR-Path-Packet-Loss on each measurement loop m_i , m_j and m_k by waiting for T_{max} to receive any of

the missed packets, this allows for a reaction ~~independant~~independent of a conservative packet loss threshold like Tmax. The idea is to judge on disconnectivity if no packet is received on all three measurement loops mi, mj and mk after the time interval the last single packet was expected to be received, if there was no prior indication of congestion.

If the spacing of packets along consecutive measurement loops Fi is IncF as defined within ~~section~~ Section 4.4, then under stable network conditions every measurement packet sent along measurement loop Fi is received, before the next measurement packet is sent along measurement loop Fj. If a measurement interval starts at Tl and none of the three measurement loops Fi, Fj and Fk received a packet within $Tl + incT = Tl + 6 * incF$, monitored Sub-Path i is disconnected. It doesn't matter, along which of the three measurement loops the first not received packet was sent (there's no order here).

$incF > \max (SR-Path-Delay-MeanCSi + d * Delta * SR-Path-Delay-StdCSi), i \text{ in } [1...6]$

With d and Delta being integer numbers as specified in ~~section~~ Section 11.4. If Fi and Fi+1 are measurement loops along which measurement packets are sent in consecutive order, this definition of incF ensures that the measurement packet sent along measurement loop Fi is received prior to sending the next measurement packet along measurement loop Fi+1 (under stable network conditions). The product $d * Delta * SR-Path-Delay-StdCSi$ allows to set the preferred tolerance for outliers. It impacts the tradeoff between speed of detection and false positive ratio. With this parameterisation, the metric ~~indications~~indicating a loss of bidirectional connectivity along Sub-Path i is defined as

either zero or one (or some logical equivalent), where LofCi=1 indicates loss of continuity along monitored Sub-Path Fi and LofCi=0 indicates successful arrival of at least one packet sent along measurement-loop Fi, Fj or Fk within incT.

Under conditions of ~~section~~ Section 4.4, if at any sliding interval incT no singleton was received along measurement-loops Fi, Fj and Fk, no more packets are forwarded in any direction of monitored sub-path SPi.

Faster detection of disconnectivity is likely possible by a different metric definition, which likely will depend on the measurement-loop delay Mi, Mj and Mk. The metric chosen above allows for a simple parametrisation. Metrics allowing for a faster determination of disconnection are not within scope of this document.

The sub-path SPi is judged to be disconnected from the earliest time, when a packet was sent but not received on any of the three sub-paths Fi, Fj or Fk. The sub-path SPi is judged to be connected, whenever a measurement packet sent along one or more of the measurement-loops Fi, Fj and Fk is received again.

F_i = send time of a packet along measurement-loop F_i
 i in $[1..6]$
 M_i = receive time of a packet sent along F_i
 $incT$ interval between two packets sent along F_i
 $incF > \max (M_i)$

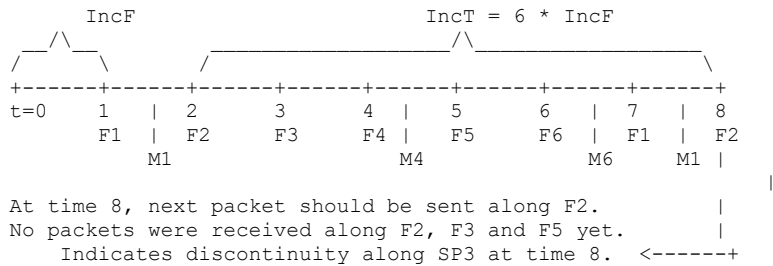


Figure 3

Illustration of the sub-path disconnectivity metric; sub-path SP3 is link L100 <-> L070 of the example network Figure 1.

Note, if F2 sent at time 2 was received at time 2 + M2, but no more packet passing SP3 afterwards, discontinuity of SP3 is indicated at time 9, when F3 is to send the next packet. Also note that discontinuity of SP3 could be indicated as early as time 6 in the example. That requires a different metric. Basing the metric definition on incT however covers all potential intervals between relevant F_i , F_j and F_k .

12. Discussion of Temporal Resolution

A loss of connectivity is detected after a temporal distance of $incT$, the time period between two packets ~~beingbeing~~ sent along the same measurement-loop F_i . $incT$ is specified as $6 \cdot incF$, where $incF$ is 2 times the largest measurement-loop delay in the absence of congestion. Hence a loss of connectivity is indicated after $12 \cdot$ the largest measurement-loop delay.

Reliable indications of lost connectivity may be possible also at smaller timescales. The specification chosen seems to be simple as well as reliable and thus defines a starting point for advanced designs offering faster reaction.

13. IANA Considerations

If standardised, the metric will require an entry in the IPPM metric registry.

14. Security Considerations

This ~~draft-document~~ specifies how to use methods specified or described within [RFC8402] and [RFC8403]. It does not introduce new or additional SR features. The security considerations of both references apply here

Commenté [MB5]: The metric registry is Specification Required.
The metric can be registered.

too.

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